

Google Cloud Maps API Cost & Capacity Report

Project: UnivoAdmin | Architecture Analysis & Cost Breakdown | Date: September 2026

EXECUTIVE SUMMARY:

Google Cloud provides a **\$200.00 USD Monthly Recurring Free Credit** (approx. **Rs. 16,600 INR**) to every active billing account. For standard UnivoAdmin field operations (searching libraries/gyms, checking client phone numbers, logging follow-ups, and GPS road distance calculation), **the net out-of-pocket cost is Rs. 0.00 (100% Free)**.

1. Case Study: The 51 Libraries Discovery Search

When an admin or field staff member searches 'library' within a 5 km radius and **51 libraries** appear on the screen:

Component	Mechanism	API Calls	Gross Cost	Net Billed
Places Text Search	Google fetches 20 places per page. 51 results require 3 pages (20 + 20 + 11).	3 requests	\$0.015 - \$0.060 (Rs. 1.25 - Rs. 5.00)	Rs. 0.00 (from free credit)
Routes Distance Matrix	Computes real driving road distance and travel time for first 25 nearest places.	1 batch call	\$0.005 (Rs. 0.40)	Rs. 0.00 (from free credit)
Total Search Cost	Combined cost for discovering 51 libraries simultaneously.	4 calls total	Rs. 2.00 - Rs. 5.50	Rs. 0.00 (FREE)

Important Note: Google does NOT charge per library found. It charges per page request. Whether 5 libraries appear or 51 libraries appear, it is billed in 20-result pages, keeping the total cost at just a few rupees.

2. Turn-by-Turn Navigation Cost: 100% Free (Rs. 0)

Does clicking 'Navigate' or 'Open in Google Maps' cost any money?

NO. It is 100% COMPLETELY FREE.

When a staff rep clicks the Navigation button on a library card, UnivoAdmin generates a standard deep link:

<https://www.google.com/maps/dir/?api=1&origin;=...&destination;=...>

This opens the Google Maps mobile app or web browser directly. Google **never charges** for opening directions or GPS navigation in Google Maps. Your field staff can click Navigation **unlimited times daily** with zero cost.

3. Monthly Free Capacity (\$200 / Rs. 16,600 Allowance)

Under the monthly free tier, the system can support the following volume without paying anything:

Operational Activity	Free Monthly Limit	Free Daily Limit	Typical Team Need
Full Discovery Searches (e.g., 5 km library/gym searches)	2,500 to 5,000 searches	80 to 160 searches / day	10 to 30 searches / day (~15% of limit)
Total Libraries Seen on Screen (Discovery results list)	75,000 to 100,000+ places	2,500 to 3,500 places / day	200 to 500 places / day (~10% of limit)
Phone & Place Details Lookup (Auto-fetch owner contact on Log Visit)	40,000+ lookups	1,300+ lookups / day	20 to 50 visits / day (< 3% of limit)
GPS Driving Navigation (Live turn-by-turn road navigation)	Unlimited (Always Rs. 0)	Unlimited (Always Rs. 0)	As needed by all staff reps

4. Google Cloud Service-by-Service Pricing Architecture

Official Google Maps Platform list rates (applicable only if the \$200 free monthly credit is exceeded):

Google API Service	Usage in UnivoAdmin	Cost per 1,000	Cost per Single Call
Places API (New) - Text Search	Finding nearby libraries & gyms within custom radius	\$5.00 to \$32.00	Rs. 0.40 to Rs. 2.60 per 20-place page
Places API - Place Details (Contact)	Auto-populating phone number and website on Log Visit	\$3.00	Rs. 0.25 per place lookup
Routes API - ComputeRouteMatrix	Exact road travel time and driving distance	\$5.00	Rs. 0.40 per batch matrix call
Geocoding API	Translating typed city / area name into coordinates	\$5.00	Rs. 0.40 per city lookup
Maps JavaScript API	Loading base map library on Field Marketing page	\$7.00	Rs. 0.58 per session load

5. Built-in Cost Optimizations Implemented in Code

To ensure zero unnecessary API charges, UnivoAdmin includes three automated safeguards:

- 1. Haversine Math (Zero API Calls):** Initial distance calculation and sorting is computed using local mathematical trigonometry (Haversine algorithm). This eliminates thousands of redundant Google distance requests.
- 2. Batch Matrix Slicing:** Driving distance matrix queries are capped at a maximum of 25 places (`validPlaces.slice(0, 25)`). Even if 100 libraries are discovered, only the 25 nearest ones query the Routes API.
- 3. On-Demand Lazy Loading:** The Google Maps JavaScript library is only loaded when entering the Field Marketing page, preventing background API initialization on login or dashboard load.

6. Advanced Best Practices & Security Setup

Follow these recommendations in your Google Cloud Console to ensure complete peace of mind:

Security / Cost Step	Recommended Action	How to Set Up
1. Set Budget Alert	Receive instant automated email alerts if usage reaches 50% or 90% of free credit.	Google Cloud Console → Billing → Budgets & Alerts → Set Target to \$150 or Rs. 1,000.
2. Restrict API Key	Lock your API key to your specific domain so external parties cannot use your quota.	APIs & Services → Credentials → Edit Key → <i>Website Restrictions (HTTP referrers)</i> .
3. Set Daily Quotas	Enforce a hard daily request ceiling (e.g. 500 searches/day) to prevent runaway scripts.	APIs & Services → Places API → Quotas → Set <i>Requests per day</i> limit.
4. Live Billing Check	Monitor exact daily usage, credits applied, and net balance in real time.	Google Cloud Console → Billing → Reports (Filtered by SKU & Service).

FINAL VERDICT:

For your business operations, **your Google API expenditure is Rs. 0.00 / month**. The \$200 free credit provides ample buffer for more than 75,000 library discoveries every month. Follow the budget alert setup above to ensure 100% cost safety.